

Аналіз мережевої активності шкідливого ПЗ

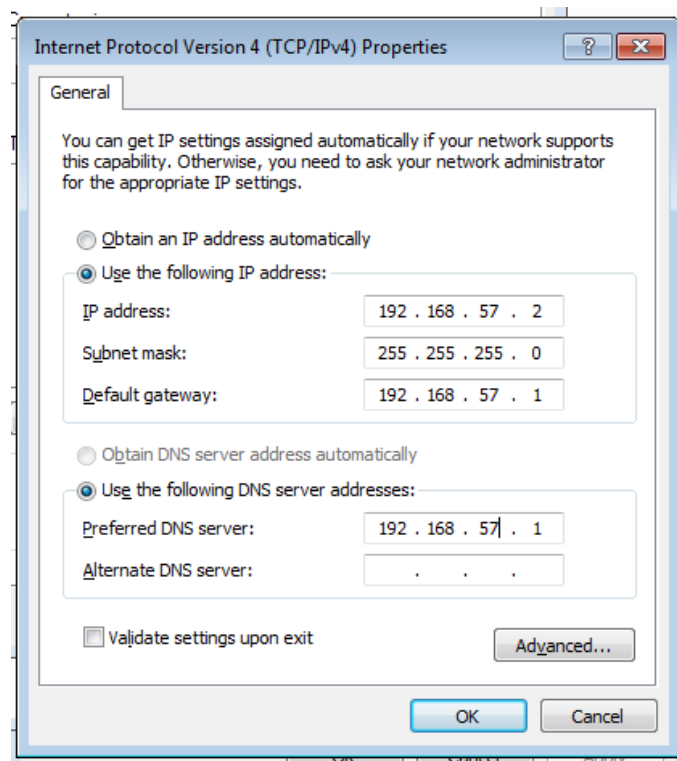
Кувавина Софія

Мета роботи: Отримати навички аналізу мережевих комунікацій ШПЗ.

Хід роботи:

1. Додайте INetSim у Cuckoo Sandbox з розділу 3.3.1. Проаналізуйте 3-5 зразків з theZoo [58].

Налаштовую статичну IP-адресу та DNS-сервер на клієнті Windows 7 для використання INetSim як фейкового DNS-сервера



хост-онлі мережа

```
Name:                vboxnet1
GUID:                786f6276-656e-4174-8000-0a0027000001
DHCP:                Disabled
IPAddress:           192.168.57.1
NetworkMask:         255.255.255.0
IPV6Address:         fe80::800:27ff:fe00:1
IPV6NetworkMaskPrefixLength: 64
HardwareAddress:     0a:00:27:00:00:01
MediumType:          Ethernet
Wireless:            No
Status:              Up
VBoxNetworkName:     HostInterfaceNetworking-vboxnet1
```

```
C:\Users\cuckoo>ping 192.168.57.1

Pinging 192.168.57.1 with 32 bytes of data:
Reply from 192.168.57.1: bytes=32 time=1ms TTL=64
Reply from 192.168.57.1: bytes=32 time<1ms TTL=64
Reply from 192.168.57.1: bytes=32 time<1ms TTL=64
Reply from 192.168.57.1: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.57.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Users\cuckoo>_
```

```
sofia@ubuntu:~$ ping 192.168.57.2
PING 192.168.57.2 (192.168.57.2) 56(84) bytes of data.
64 bytes from 192.168.57.2: icmp_seq=1 ttl=128 time=33.6 ms
64 bytes from 192.168.57.2: icmp_seq=2 ttl=128 time=0.722 ms
64 bytes from 192.168.57.2: icmp_seq=3 ttl=128 time=0.662 ms
64 bytes from 192.168.57.2: icmp_seq=4 ttl=128 time=1.17 ms
^C
--- 192.168.57.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3032ms
rtt min/avg/max/mdev = 0.662/9.058/33.678/14.215 ms
```

Встанавливаю INetSim

```
sofia@ubuntu:~$ wget http://www.inetsim.org/debian/binary/inetsim_1.3.2-1_all.deb
--2025-11-15 03:23:19-- http://www.inetsim.org/debian/binary/inetsim_1.3.2-1_all.deb
Resolving www.inetsim.org (www.inetsim.org)... 85.214.152.164
Connecting to www.inetsim.org (www.inetsim.org)|85.214.152.164|:80... connected.
HTTP request sent, awaiting response... 200 OK
Length: 231292 (226K) [application/vnd.debian.binary-package]
Saving to: 'inetsim_1.3.2-1_all.deb'

inetsim_1.3.2-1_all 100%[=====] 225.87K 1001KB/s in 0.2s

2025-11-15 03:23:20 (1001 KB/s) - 'inetsim_1.3.2-1_all.deb' saved [231292/231292]
```

Змінимо налаштування у файлах конфігурацій inetsim.conf,

```
GNU nano 2.5.3 File: /etc/inetsim/inetsim.conf

start_service daytime_tcp
start_service daytime_udp
start_service echo_tcp
start_service echo_udp
start_service discard_tcp
start_service discard_udp
start_service quotd_tcp
start_service quotd_udp
start_service chargen_tcp
start_service chargen_udp
start_service dummy_tcp
start_service dummy_udp

#####
# service_bind_address
#
# IP address to bind services to
#
# Syntax: service_bind_address <IP address>
#
# Default: 127.0.0.1
#
service_bind_address 192.168.57.1

#####

#####
# dns_default_ip
#
# Default IP address to return with DNS replies
#
# Syntax: dns_default_ip <IP address>
#
# Default: 127.0.0.1
#
dns_default_ip 192.168.57.1

#####
```

```
GNU nano 2.5.3 File: /home/sofiia/.cuckoo/conf/machines.conf

[cuckoo1]
label = cuckoo1
platform = windows
ip = 192.168.57.2
interface = vboxnet1
snapshot = clean1
osprofile = win7
```

Запускаю inetsim і бачимо реєстрацію емульованих мережесервісів

```
sofia@ubuntu:~$ sudo inetsim --bind-address 192.168.57.1
[sudo] password for sofia:
INetSim 1.3.2 (2020-05-19) by Matthias Eckert & Thomas Hungenberg
Using log directory:      /var/log/inetsim/
Using data directory:     /var/lib/inetsim/
Using report directory:   /var/log/inetsim/report/
Using configuration file: /etc/inetsim/inetsim.conf
Parsing configuration file.
Configuration file parsed successfully.
=== INetSim main process started (PID 34459) ===
Session ID:      34459
Listening on:    192.168.57.1
Real Date/Time:  2025-11-15 05:13:54
Fake Date/Time: 2025-11-15 05:13:54 (Delta: 0 seconds)
Forking services...
* http_80_tcp - started (PID 34462)
* dns_53_tcp_udp - started (PID 34461)
* pop3_110_tcp - started (PID 34466)
* ftp_21_tcp - started (PID 34468)
* https_443_tcp - started (PID 34463)
* ident_113_tcp - started (PID 34474)
* tftp_69_udp - started (PID 34470)
* smtp_25_tcp - started (PID 34464)
* smtps_465_tcp - started (PID 34465)
* syslog_514_udp - started (PID 34475)
* ntp_123_udp - started (PID 34472)
* irc_6667_tcp - started (PID 34471)
* ftps_990_tcp - started (PID 34469)
* time_37_tcp - started (PID 34476)
* finger_79_tcp - started (PID 34473)
* time_37_udp - started (PID 34477)
* pop3s_995_tcp - started (PID 34467)
* chargen_19_tcp - started (PID 34486)
* dummy_1_udp - started (PID 34489)
* quotd_17_udp - started (PID 34485)
* quotd_17_tcp - started (PID 34484)
* echo_7_udp - started (PID 34481)
* daytime_13_tcp - started (PID 34478)
* echo_7_tcp - started (PID 34480)
* dummy_1_tcp - started (PID 34488)
* discard_9_udp - started (PID 34483)
* discard_9_tcp - started (PID 34482)
* daytime_13_udp - started (PID 34479)
* chargen_19_udp - started (PID 34487)
done.
Simulation running.
```

```
sofia@ubuntu:~$ service inetsim status
● inetsim.service - LSB: start and stop the internet simulation
   Loaded: loaded (/etc/init.d/inetsim; bad; vendor preset: enabled)
   Active: active (exited) since Sat 2025-11-15 03:23:53 PST; 1h 51min ago
     Docs: man:systemd-sysv-generator(8)

Nov 15 03:23:51 ubuntu systemd[1]: Starting LSB: start and stop the internet simulation...
Nov 15 03:23:51 ubuntu inetsim[32807]: * Starting Internet Service Simulation Suite inetsim
Nov 15 03:23:53 ubuntu inetsim[32807]: ...done.
Nov 15 03:23:53 ubuntu systemd[1]: Started LSB: start and stop the internet simulation.
```

1. Somoto

File 7ZipSetup.exe			
Summary		Download	Resubmit sample
Size	158.2KB		
Type	PE32 executable (GUI) Intel 80386 (stripped to external PDB), for MS Windows, Nullsoft Installer self-extracting archive		
MD5	02e0b78e2876087f678f070ed60e4c30		
SHA1	28ab8945612608716ca3959061ce79b92b9c5f41		
SHA256	ddf2542dc5ac74a98d5ee9e55497572104d6c880aad9137caf884d10ca5953ce		
SHA512	Show SHA512		
CRC32	2BE11914		
ssdeep	None		
Yara	None matched		

Analysis					
Category	Started	Completed	Duration	Routing	Logs
FILE	Nov. 19, 2025, 11:59 a.m.	Nov. 19, 2025, 12:02 p.m.	214 seconds	none	Show Analyzer Log Show Cuckoo Log

1	Checks amount of memory in system, this can be used to detect virtual machines that have a low amount of memory available (1 event)	>
1	The executable contains unknown PE section names indicative of a packer (could be a false positive) (1 event)	>
1	Performs some HTTP requests (1 event)	▼
request	GET http://bi.blsrv.com/awde7zip19538/7zip/ce6918312bbbd24eb00bfe46c7654d55?v=2.0&uid=B3B4AB37F5E0FED3B4A10DCF2DECE21E	localhost:8000/analysis/11/summary/#signature allocates

The executable contains unknown P.E. section names indicative of a packer (could be a false positive) (1 event)				
Performs some HTTP requests (1 event)				
request	GET http://bi.bisrv.com/awde7zip19538/7zip/ce6918312bbbd24eb00bfe46c7654d557v=2.0&muid=B3B4AB37F5E0FED3B4A10DCF2DECE21E			
Allocates read-write-execute memory (usually to unpack itself) (1 event)				
Time & API	Arguments	Status	Return	Repeated
NTAllocateVirtualMemory Nov. 19, 2025, 12:01 p.m.	process_identifier: 2912 region_size: 4096 stack_dep_bypass: 0 stack_pivoted: 0 heap_dep_bypass: 0 protection: 64 (PAGE_EXECUTE_READWRITE) base_address: 0x002a0000 allocation_type: 4096 (MEM_COMMIT) process_handle: 0xffffffff	1	0	0
Creates executable files on the filesystem (1 event)				
file	C:\Users\cuckoo\AppData\Local\Temp\biclient.exe			
Drops an executable to the user AppData folder (1 event)				
file	C:\Users\cuckoo\AppData\Local\Temp\biclient.exe			
Checks adapter addresses which can be used to detect virtual network interfaces (1 event)				
Time & API	Arguments	Status	Return	Repeated
GetAdaptersAddresses Nov. 19, 2025, 12:01 p.m.	flags: 0 family: 0		111	0
Deletes executed files from disk (1 event)				
file	C:\Users\cuckoo\AppData\Local\Temp\biclient.exe			

file		C:\Users\cuckoo\AppData\Local\Temp\bi client.exe		
✖ Disables proxy possibly for traffic interception (1 event)				
Time & API	Arguments	Status	Return	Repeated
RegSetValueExA Nov. 19, 2025, 12:01 p.m.	key_handle: 0x00000320 regkey_r: ProxyEnable reg_type: 4 (REG_DWORD) value: 0 regkey: HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Internet Settings\ProxyEnable	1	0	0
✖ Sets or modifies WPAD proxy autoconfiguration file for traffic interception (15 events)				
Time & API	Arguments	Status	Return	Repeated
RegSetValueExA Nov. 19, 2025, 12:01 p.m.	key_handle: 0x0000043c regkey_r: WpadDecisionReason reg_type: 4 (REG_DWORD) value: 1 regkey: HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Internet Settings\Wpad\{71C32B11-8CDF-4EB7-9FC3-8A59129A5B9E}\WpadDecisionReason	1	0	0
RegSetValueExA Nov. 19, 2025, 12:01 p.m.	key_handle: 0x0000043c regkey_r: WpadDecisionTime reg_type: 3 (REG_BINARY) value: ªÀ ÈÿÜ regkey: HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Internet Settings\Wpad\{71C32B11-8CDF-4EB7-9FC3-8A59129A5B9E}\WpadDecisionTime	1	0	0
RegSetValueExA Nov. 19, 2025, 12:01 p.m.	key_handle: 0x0000043c regkey_r: WpadDecision reg_type: 4 (REG_DWORD) value: 3 regkey: HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Internet Settings\Wpad\{71C32B11-8CDF-4EB7-9FC3-8A59129A5B9E}\WpadDecision	1	0	0
RegSetValueExW Nov. 19, 2025, 12:01 p.m.	key_handle: 0x0000043c regkey_r: WpadNetworkName reg_type: 1 (REG_SZ) value: Network regkey: HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Internet Settings\Wpad\{71C32B11-8CDF-4EB7-9FC3-8A59129A5B9E}\WpadNetworkName	1	0	0
RegSetValueExA Nov. 19, 2025, 12:01 p.m.	key_handle: 0x0000044c regkey_r: WpadDecisionReason reg_type: 4 (REG_DWORD) value: 1	1	0	0
RegSetValueExW Nov. 19, 2025, 12:01 p.m.				
RegSetValueExW Nov. 19, 2025, 12:01 p.m.	key_handle: 0x00000468 regkey_r: WpadNetworkName reg_type: 1 (REG_SZ) value: Network regkey: HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Internet Settings\Wpad\{71C32B11-8CDF-4EB7-9FC3-8A59129A5B9E}\WpadNetworkName	1	0	0
RegSetValueExA Nov. 19, 2025, 12:01 p.m.	key_handle: 0x0000046c regkey_r: WpadDecisionReason reg_type: 4 (REG_DWORD) value: 1 regkey: HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Internet Settings\Wpad\0a-00-27-00-01\WpadDecisionReason	1	0	0
RegSetValueExA Nov. 19, 2025, 12:01 p.m.	key_handle: 0x0000046c regkey_r: WpadDecisionTime reg_type: 3 (REG_BINARY) value: ªÀ ÈÿÜ regkey: HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Internet Settings\Wpad\0a-00-27-00-01\WpadDecisionTime	1	0	0
RegSetValueExA Nov. 19, 2025, 12:01 p.m.	key_handle: 0x0000046c regkey_r: WpadDecision reg_type: 4 (REG_DWORD) value: 0 regkey: HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Internet Settings\Wpad\0a-00-27-00-01\WpadDecision	1	0	0
✖ Connects to an IP address that is no longer responding to requests (legitimate services will remain up-and-running usually) (1 event)				
dead_host	10.13.37.1:80			
📷 Screenshots				
No screenshots available.				
Name		Response	Post-Analysis Lookup	
bi.bisrv.com		A → 192.168.57.1		
IP Address		Status	Action	
		No hosts contacted.		

Під час аналізу зразка SOMOTO було зафіксовано активність рекламного інсталера. Програма створює тимчасові файли у AppData, перевіряє параметри системи та виконує мережеві запити. Видно HTTP-запит на домен bi.bisrv.com, який у нашому середовищі був перенаправлений INetSim на 192.168.57.1, тому зовнішній сервер фактично не контактувався

2. teslacrypt

cuckoo

Dashboard
 Recent
 Pending
 Search

Summary

File `ed01ebfbc9eb5bbea545af4d01bf5f1071661840480439c6e5babe8e080e41aa.exe`

Summary	
Size	3.4MB
Type	PE32 executable (GUI) Intel 80386, for MS Windows
MD5	84c82835a5d21bbc7f5a61706d8ab549
SHA1	5ff465afaabcdf0150d1a3ab2c2e74f3a4426467
SHA256	ed01ebfbc9eb5bbea545af4d01bf5f1071661840480439c6e5babe8e080e41aa
SHA512	Show SHA512
CRC32	4022FCAA
ssdeep	None
Yara	None matched

Information on Execution

Analysis					
Category	Started	Completed	Duration	Routing	Logs
FILE	Nov. 15, 2025, 2:17 p.m.	Nov. 15, 2025, 2:20 p.m.	202 seconds	none	Show Analyzer Log Show Cuckoo Log

Signatures

- Queries for the computername (2 events)
- Command line console output was observed (2 events)

`ed01ebfbc9eb5bbea545af4d01bf5f1071661840480439c6e5babe8e080e41aa.exe`

Score

This file is **very suspicious**, with a score of **14.2 out of 10!**

Please notice: The scoring system is currently still in development and should be considered an alpha feature.

Feedback

Expecting different results? Send us this analysis and we will inspect it. [Click here](#)

localhost:8000/analysis/10/summary/#signature_generates_crypto_key

Checks whether any human activity is being performed by constantly checking whether the foreground window changed	>
A process attempted to delay the analysis task. (1 event)	>
Creates (office) documents on the filesystem (5 events)	>
Creates executable files on the filesystem (5 events)	>
Creates hidden or system file (50 out of 936 events)	>
Creates a shortcut to an executable file (1 event)	>
Drops an executable to the user AppData folder (3 events)	>
A process created a hidden window (9 events)	>
Changes read-write memory protection to read-execute (probably to avoid detection when setting all RWX flags at the same time) (1 event)	>
The binary likely contains encrypted or compressed data indicative of a packer (2 events)	>
Uses Windows utilities for basic Windows functionality (1 event)	>
Attempts to identify installed AV products by installation directory (6 events)	>
file	Z:\pafish-repo\pafish\sandboxie.c
file	Z:\pafish-repo\pafish\sandboxie.h.WNCRYT
file	Z:\pafish-repo\pafish\sandboxie.c.WNCRYT
file	Z:\pafish-repo\pafish\sandboxie.h
file	Z:\pafish-repo\pafish\sandboxie.h.WNCRY
file	Z:\pafish-repo\pafish\sandboxie.c.WNCRY
Putty Files, Registry Keys and/or Mutexes Detected	>

localhost:8000/analysis/10/summary/#signature_detect_putty

⚠ Appends a known WannaCry ransomware file extension to files that have been encrypted (50 out of 4243 events)				>
⚠ Deletes a large number of files from the system indicative of ransomware, wiper malware or system destruction (50 out of 4064 events)				>
⚠ Writes a potential ransom message to disk (1 event)				▼
Time & API	Arguments	Status	Return	Repeated
NtWriteFile Nov. 15, 2025, 2:17 a.m. ⓘ	buffer:Q: What's wrong with my files? A: Oops, your important files are encrypted. It means you will not be able to access them anymore until they are decrypted. If you follow our instructions, we guarantee that you can decrypt all your files quickly and safely! Let's start decrypting! Q: What do I do? A: First, you need to pay service fees for the decryption. Please send \$300 worth of bitcoin to this bitcoin address: 115p7UMMngo1pMvKpHjicRdfJNXj6LrLn Next, please find an application file named "@WanaDecryptor.exe". It is the decrypt software. Run and follow the instructions! (You may need to disable your antivirus for a while.) Q: How can I trust? A: Don't worry about decryption. We will decrypt your files surely because nobody will trust us if we cheat users. * If you need our assistance, send a message by clicking «Contact Us» on the decryptor window. offset: 0 file_handle: 0x0000010c filepath: C:\Users\cuckoo\AppData\Local\Temp\@Please_Read_Me@.txt	1	0	0
⚠ Creates and runs a batch file to remove the original binary (1 event)				▼
file	Oeefeaafd97895d_get externals.bat			
⚠ Uses suspicious command line tools or Windows utilities (1 event)				▼
cmdline	icacls . /grant Everyone:F /T /C /Q			
⚠ Connects to an IP address that is no longer responding to requests (legitimate services will remain up-and-running usually) (1 event)				▼
dead_host	10.13.37.1:80			
⚠ Drops 513 unknown file mime types indicative of ransomware writing encrypted files back to disk (50 out of 512 events)				>
⚠ Performs 4175 file moves indicative of a ransomware file encryption process (50 out of 4175 events)				>
⚠ Appends a new file extension or content to 4175 files indicative of a ransomware file encryption process (50 out of 4175 events)				>
📷 Screenshots				

Він створював багато файлів, у тому числі приховані та .exe, міняв права доступу і намагався обійти аналіз (перевіряв активність користувача, робив затримки). Програма шифрувала велику кількість файлів. Також вона створила текстовий файл “Please_Read_Me@.txt”

3. artemis

Summary

InstalIBC201401.exe

File InstalIBC201401.exe

Summary

Size

12.8MB

Type

PE32 executable (GUI) Intel 80386, for MS Windows

MD5

caff801a280d42dbd1ad6b1266d3c43a

SHA1

08b9f5874ad1dc3ee1093c9cd08737645f33f13f

SHA256

834d1dbfab8330ea5f1844f6e905ed0ac19d1033ee9a9f1122ad2051c56783dc

SHA512

Show SHA512

CRC32

6AC677AF

ssdeep

None

PDB Path

D:\Dev\Tin9\InstallDir\vc80-win32u\Loader.pdb

Yara

None matched

Download

Resubmit sample

Score

This file is **very suspicious**, with a score of **9.4 out of 10!**

Please notice:

The scoring system is currently still in development and should be considered an alpha feature.

Feedback

Expecting different results? Send us this analysis and we will inspect it. [Click here](#)

Information on Execution

Analysis

Category	Started	Completed	Duration	Routing	Logs
FILE	Nov. 19, 2025, 12:16 p.m.	Nov. 19, 2025, 12:20 p.m.	216 seconds	none	Show Analyzer Log Show Cuckoo Log

Signatures

Queries for the computername (9 events)

>

Dashboard

Recent

Pending

Search

Submit

Import

Signatures

Queries for the computername (9 events)

This executable has a PDB path (1 event)

Checks amount of memory in system, this can be used to detect virtual machines that have a low amount of memory available (1 event)

The executable contains unknown PE section names indicative of a packer (could be a false positive) (2 events)

One or more processes crashed (15 events)

Allocates read-write-execute memory (usually to unpack itself) (50 out of 85 events)

Checks whether any human activity is being performed by constantly checking whether the foreground window changed

Queries the disk size which could be used to detect virtual machine with small fixed size or dynamic allocation (6 events)

Creates executable files on the filesystem (8 events)

Creates hidden or system file (16 events)

Creates a shortcut to an executable file (17 events)

Creates a suspicious process (31 events)

Drops a binary and executes it (1 event)

Drops an executable to the user AppData folder (3 events)

Changes read-write memory protection to read-execute (probably to avoid detection when setting all RWX flags at the same time) (1 event)

The binary likely contains encrypted or compressed data indicative of a packer (3 events)

Checks for the Locally Unique Identifier on the system for a suspicious privilege (8 events)

Changes read-write memory protection to read-execute (probably to avoid detection when setting all RWX flags at the same time) (1 event)

The binary likely contains encrypted or compressed data indicative of a packer (3 events)

Checks for the Locally Unique Identifier on the system for a suspicious privilege (8 events)

Queries for potentially installed applications (27 events)

Resumed a suspended thread in a remote process potentially indicative of process injection (2 events)

Time & API	Arguments	Status	Return	Repeated
Process injection	Process 1096 resumed a thread in remote process 1640			
NtResumeThread	thread_handle: 0x0000000000000ba0 suspend_count: 1 process_idenfier: 1640	1	0	0

Performs 106 file moves indicative of a ransomware file encryption process (50 out of 106 events)

Appends a new file extension or content to 106 files indicative of a ransomware file encryption process (50 out of 106 events)

Connects to an IP address that is no longer responding to requests (legitimate services will remain up-and-running usually) (1 event)

dead_host

10.13.37.180

Screenshots

No screenshots available.

Name	Response	Post-Analysis Lookup	IP Address	Status	Action
		No hosts contacted.			No hosts contacted.

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перевіряє параметри системи, створює приховані та системні файли, робить багато підозрілих змін на диску, крашить процеси і виконує процес-ін'єкцію. Також спостерігаються масові операції над файлами, що схоже на поведінку шифрувальника. Є спроба мережевого звернення до 10.13.37.1, яке перехопив INetSim

2. Розгорніть OpenVPN за допомогою openvpn-install [102], робота за протоколом TCP. На стороні клієнта встановіть з'єднання з OpenVPN сервером через HTTP проксі. Проксі можна отримати за допомогою fetch-some-proxies [103] або онлайн сервісів

Встановлюю та налаштовую OpenVPN

```
(kali@kali)~[/lab5]
$ sudo apt update
sudo apt install -y curl openvpn git python3
[sudo] password for kali:
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/main arm64 Packages [20.8 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Packages [21.0 MB]
Get:4 http://kali.download/kali kali-rolling/main arm64 Contents (deb) [49.1 MB]
Get:5 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [52.6 MB]
Get:6 http://kali.download/kali kali-rolling/contrib amd64 Packages [114 kB]
Get:7 http://kali.download/kali kali-rolling/contrib arm64 Packages [99.8 kB]
Get:8 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [259 kB]
Get:9 http://kali.download/kali kali-rolling/contrib arm64 Contents (deb) [179 kB]
Get:10 http://kali.download/kali kali-rolling/non-free arm64 Packages [146 kB]
Get:11 http://kali.download/kali kali-rolling/non-free amd64 Packages [186 kB]
Get:12 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [893 kB]
Get:13 http://kali.download/kali kali-rolling/non-free arm64 Contents (deb) [849 kB]
Fetched 146 MB in 22s (6.595 kB/s)
```

Надаю права на виконання та налаштовую OpenVPN-сервер за допомогою openvpn-install

```
(kali@kali)~[/lab5]
$ chmod +x openvpn-install.sh

(kali@kali)~[/lab5]
$ sudo ./openvpn-install.sh
Welcome to the OpenVPN installer!
The git repository is available at: https://github.com/angristan/openvpn-install

I need to ask you a few questions before starting the setup.
You can leave the default options and just press enter if you are okay with them.

I need to know the IPv4 address of the network interface you want OpenVPN listening to.
Unless your server is behind NAT, it should be your public IPv4 address.
IP address: 192.168.145.129

It seems this server is behind NAT. What is its public IPv4 address or hostname?
We need it for the clients to connect to the server.
Public IPv4 address or hostname: 77.47.170.240

Checking for IPv6 connectivity ...

Your host does not appear to have IPv6 connectivity.

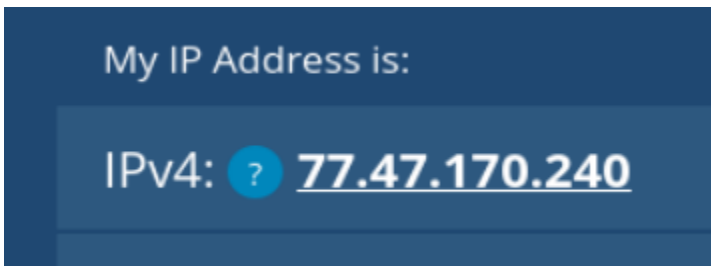
Do you want to enable IPv6 support (NAT)? [y/n]: n

What port do you want OpenVPN to listen to?
1) Default: 1194
2) Custom
3) Random [49152-65535]
Port choice [1-3]: 1

What protocol do you want OpenVPN to use?
UDP is faster. Unless it is not available, you shouldn't use TCP.
1) UDP
2) TCP
Protocol [1-2]: 2

What DNS resolvers do you want to use with the VPN?
1) Current system resolvers (from /etc/resolv.conf)
2) Self-hosted DNS Resolver (Unbound)
3) Cloudflare (Anycast: worldwide)
4) Quad9 (Anycast: worldwide)
```

Перевірка публічної IP-адреси Kali Linux до підключення VPN-тунелю



Заміняю публічну IP на локальну IP моєї Kali-vm:

```
GNU nano 8.4 /home/kali/client1.ovpn *
client
proto tcp-client
remote 192.168.145.129 1194
dev tun
resolv-retry infinite
nobind
persist-key
persist-tun
remote-cert-tls server
verify-x509-name server_xkU1eDI9M1jViiUV name
auth SHA256
auth-nocache
cipher AES-128-GCM
tls-client
```

Підключаємося до проксі, клонуючи файл з репозиторію

Запускаємо файл fetch.py для пошуку проксі, але у моїй версії інструмента скрипт повертав лише SOCKS4/5-проксі, які несумісні з директивою http-проху OpenVPN, оскільки OpenVPN підтримує тільки HTTP/HTTPS-проксі, а онлацн -проксі блокує мережа

```
(kali@kali)-[~/lab5]
$ git clone https://github.com/stamparm/fetch-some-proxies
Cloning into 'fetch-some-proxies'...
remote: Enumerating objects: 192, done.
remote: Counting objects: 100% (15/15), done.
remote: Compressing objects: 100% (11/11), done.
remote: Total 192 (delta 8), reused 11 (delta 4), pack-reused 177 (from 1)
Receiving objects: 100% (192/192), 40.46 KiB | 390.00 KiB/s, done.
Resolving deltas: 100% (68/68), done.

(kali@kali)-[~/lab5]
$ cd fetch-some-proxies

(kali@kali)-[~/lab5/fetch-some-proxies]
$ python3 fetch.py
+-----+
f e e t c h | s o m e | p r o x i e s | ← v3.2.4
+-----+

[i] initial testing...
[i] retrieving list of proxies...
[i] testing 921 proxies (20 threads)...

socks5://184.178.172.3:4145 # latency: 3.22 sec; country: United States; anonymity: elite (high)
socks4://200.41.182.243:4145 # latency: 2.38 sec; country: Argentina; anonymity: elite (high)
socks4://184.178.172.14:4145 # latency: 2.53 sec; country: United States; anonymity: elite (high)
socks4://184.178.172.18:15280 # latency: 2.71 sec; country: United States; anonymity: elite (high)
socks4://184.178.172.13:15311 # latency: 1.77 sec; country: United States; anonymity: elite (high)
socks5://174.64.199.79:4145 # latency: 2.75 sec; country: United States; anonymity: elite (high)
socks4://184.178.172.17:4145 # latency: 8.34 sec; country: United States; anonymity: elite (high)
socks5://174.75.211.222:4145 # latency: 6.79 sec; country: United States; anonymity: elite (high)
socks4://184.178.172.25:15291 # latency: 1.94 sec; country: United States; anonymity: elite (high)
-
[i] done
```


Я вирішила Поставити тінуроху як локальний HTTP-проксі

Встановлюю та вмикаю

тінуроху працює на порту 8888

```
(kali@kali)-[~/lab5]
$ sudo apt install tinyproxy -y

The following packages were automatically installed and are no longer required:
  cpp-14-arm-linux-gnueabi      gcc-15-cross-base-ports  libgcc-14-dev-armel-cross  libsigsegv2
  cpp-14-i686-linux-gnu        libasan8-armel-cross    libgcc-14-dev-i386-cross   libstdc++-14-dev-arm64-cross
  cpp-15-arm-linux-gnueabi      libasan8-i386-cross     libgcc-15-dev-armel-cross  libstdc++-14-dev-armel-cross
  cpp-arm-linux-gnueabi        libatomic1-armel-cross  libgcc-s1-armel-cross     libstdc++-14-dev-i386-cross
  cpp-i686-linux-gnu          libatomic1-i386-cross   libgcc-s1-i386-cross      libstdc++-15-dev-armel-cross
  gcc-14-arm-linux-gnueabi-base libc6-armel-cross       libgomp1-armel-cross      libstdc++6-armel-cross
  gcc-14-cross-base-ports      libc6-dev-armel-cross   libgomp1-i386-cross       libstdc++6-i386-cross
  gcc-14-i686-linux-gnu-base  libc6-dev-i386-cross    libitm1-i386-cross       libubsan1-armel-cross
  gcc-15-arm-linux-gnueabi-base libc6-i386-cross        libquadmath0-i386-cross  libubsan1-i386-cross
Use 'sudo apt autoremove' to remove them.

Installing:
  tinyproxy
```

```
(kali@kali)-[~/lab5]
$ sudo systemctl enable tinyproxy
$ sudo systemctl start tinyproxy

Synchronizing state of tinyproxy.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable tinyproxy
Created symlink '/etc/systemd/system/multi-user.target.wants/tinyproxy.service' → '/usr/lib/systemd/system/tinyproxy.service'.

(kali@kali)-[~/lab5]
$ curl -x 127.0.0.1:8888 http://example.com

<!doctype html><html lang="en"><head><title>Example Domain</title><meta name="viewport" content="width=device-width, initial-scale=1"><style>body{background:#eee;width:60vw;margin:15vh auto;font-family:system-ui,sans-serif}h1{font-size:1.5em}div{opacity:0.8}a:link,a:visited{color:#348}</style><body><div><h1>Example Domain</h1><p>This domain is for use in documentation examples without needing permission. Avoid use in operations.<p><a href="https://iana.org/domains/example">Learn more</a></div></body></html>
```

Додаємо проксі

```
GNU nano 8.4 /home/kali/client1.ovpn
client
proto tcp-client
remote 192.168.145.129 1194
http-proxy 127.0.0.1 8888
http-proxy-retry

dev tun
resolv-retry infinite
nobind
persist-key
persist-tun
remote-cert-tls server
```

Запускаємо OpenVPN-клієнт: показано встановлення TLS-з'єднання із сервером, обмін ключами та завершення ініціалізації (Initialization Sequence Completed)

```

(kali@kali)-[~/lab5]
$ sudo openvpn --config /home/kali/client1.ovpn

2025-11-16 16:15:14 DEPRECATED OPTION: http-proxy-retry and socks-proxy-retry: In OpenVPN 2.4 proxy connection retries are handled like regular con
nections. Use connect-retry-max 1 to get a similar behavior as before.
2025-11-16 16:15:14 Unrecognized option or missing or extra parameter(s) in /home/kali/client1.ovpn:21: block-outside-dns (2.6.15)
2025-11-16 16:15:14 Note: --http-proxy disables data channel offload.
2025-11-16 16:15:14 OpenVPN 2.6.15 x86_64-pc-linux-gnu [SSL (OpenSSL)] [LZO] [LZ4] [EPOLL] [PKCS11] [MH/TKINFO] [AEAD] [DCO]
2025-11-16 16:15:14 Library versions: OpenSSL 3.5.4 30 Sep 2025, LZO 2.10
2025-11-16 16:15:14 DCO version: N/A
2025-11-16 16:15:14 TCP/UDP: Preserving recently used remote address: [AF_INET]127.0.0.1:8888
2025-11-16 16:15:14 Socket Buffers: R=[131072→131072] S=[16384→16384]
2025-11-16 16:15:14 Attempting to establish TCP connection with [AF_INET]127.0.0.1:8888
2025-11-16 16:15:14 TCP connection established with [AF_INET]127.0.0.1:8888
2025-11-16 16:15:14 Send to HTTP proxy: 'CONNECT 192.168.145.129:1194 HTTP/1.0'
2025-11-16 16:15:14 Send to HTTP proxy: 'Host: 192.168.145.129'
2025-11-16 16:15:14 HTTP proxy returned: 'HTTP/1.0 200 Connection established'
2025-11-16 16:15:16 TCPv4_CLIENT link local: (not bound)
2025-11-16 16:15:16 TCPv4_CLIENT link remote: [AF_INET]127.0.0.1:8888
2025-11-16 16:15:16 TLS: Initial packet from [AF_INET]127.0.0.1:8888, sid=a3504dbb 2e09f15b
2025-11-16 16:15:16 VERIFY OK: depth=1, CN=cn_zMCPzwt2PIDOp06k
2025-11-16 16:15:16 VERIFY KU OK
2025-11-16 16:15:16 Validating certificate extended key usage
2025-11-16 16:15:16 ++ Certificate has EKU (str) TLS Web Server Authentication, expects TLS Web Server Authentication
2025-11-16 16:15:16 VERIFY EKU OK
2025-11-16 16:15:16 VERIFY X509NAME OK: CN=server_xkU1eDI9M1jVIiUV
2025-11-16 16:15:16 VERIFY OK: depth=0, CN=server_xkU1eDI9M1jVIiUV
2025-11-16 16:15:17 Control Channel: TLSv1.3, cipher TLSv1.3 TLS_AES_256_GCM_SHA384, peer certificate: 256 bits ECP256v1, signature: ecdsa-with
-SHA256
2025-11-16 16:15:17 [server_xkU1eDI9M1jVIiUV] Peer Connection Initiated with [AF_INET]127.0.0.1:8888
2025-11-16 16:15:17 TLS: move_session: dest=TM_ACTIVE src=TM_INITIAL reinit_src=1
2025-11-16 16:15:17 tls_multi_process: initial untrusted session promoted to trusted
2025-11-16 16:15:17 PUSH: Received control message: 'PUSH_REPLY,dhcp-option DNS 1.0.0.1,dhcp-option DNS 1.1.1.1,redirect-gateway def1 bypass-dhcp,r
oute-gateway 10.8.0.1,topology subnet,ping 10,ping-restart 120,ifconfig 10.8.0.2 255.255.255.0,peer-id 0,cipher AES-128-GCM,protocol-flags cc-exit
tls-ekm dyn-tls-crypt,tun-mtu 1500'
2025-11-16 16:15:17 OPTIONS IMPORT: --ifconfig/up options modified
2025-11-16 16:15:17 OPTIONS IMPORT: route options modified
2025-11-16 16:15:17 OPTIONS IMPORT: route-related options modified
2025-11-16 16:15:17 OPTIONS IMPORT: --ip-win32 and/or --dhcp-option options modified
2025-11-16 16:15:17 OPTIONS IMPORT: tun-mtu set to 1500
2025-11-16 16:15:17 net_route_v4_best_gw query: dst 0.0.0.0

```

Після підключення VPN Клієнт отримує VPN-адресу:
10.8.0.2/24 (інтерфейс tun1)

```

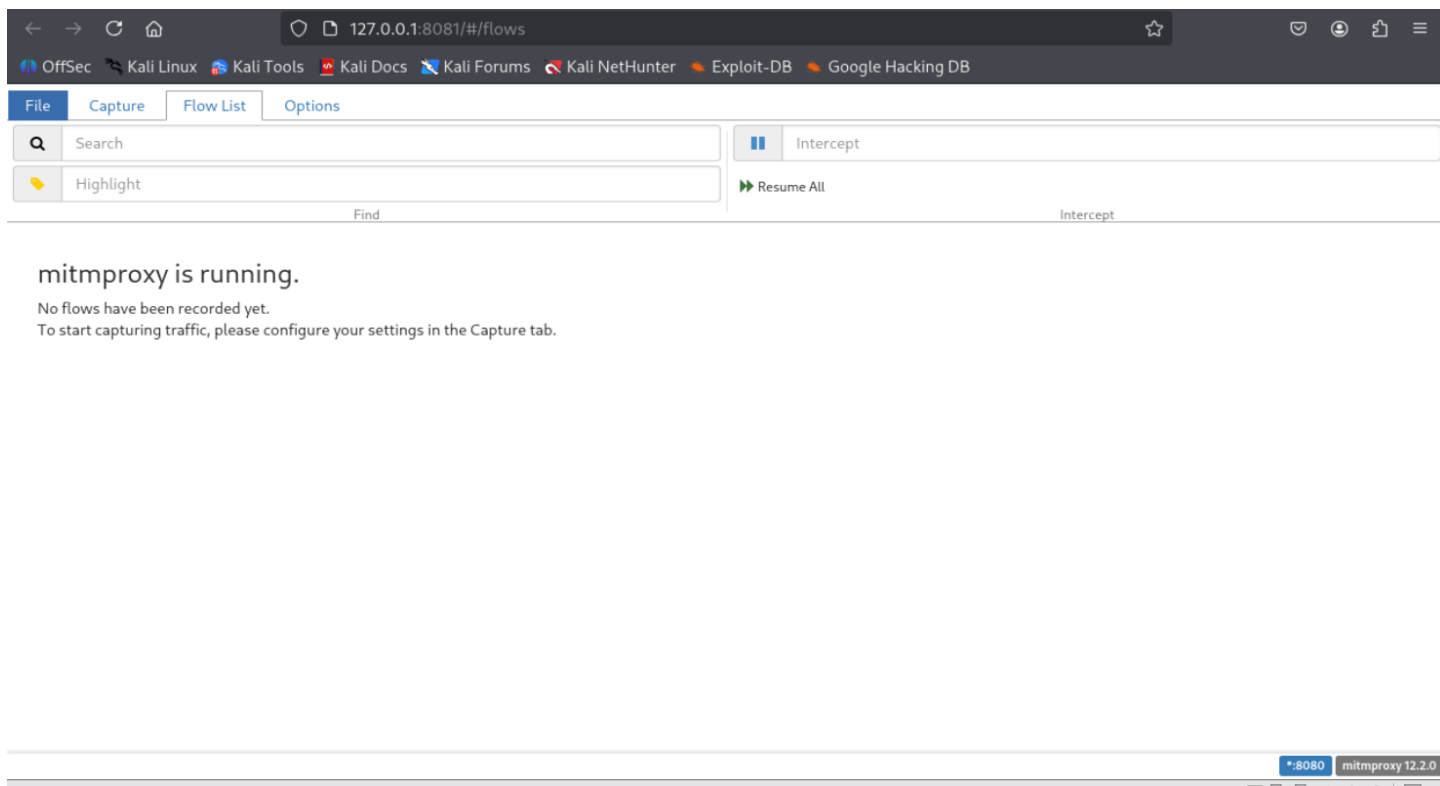
(kali@kali)-[~]
$ ifconfig tun1

tun1: flags=4305<UP,POINTOPOINT,RUNNING,NOARP,MULTICAST> mtu 1500
    inet 10.8.0.2 netmask 255.255.255.0 destination 10.8.0.2
    inet6 fe80::db77:484:434a:c3f3 prefixlen 64 scopeid 0x20<link>
    unspec 00-00-00-00-00-00-00-00-00-00-00-00-00-00-00 txqueuelen 500 (UNSPEC)
    RX packets 1 bytes 48 (48.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 3 bytes 144 (144.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

```

3. Додайте сертифікат CA mitmproxу у список довірених на клієнті (розділ 5.3.4). Проаналізуйте трафік Вашого зразку з лабораторної роботи 4

Запуск mitmproxу

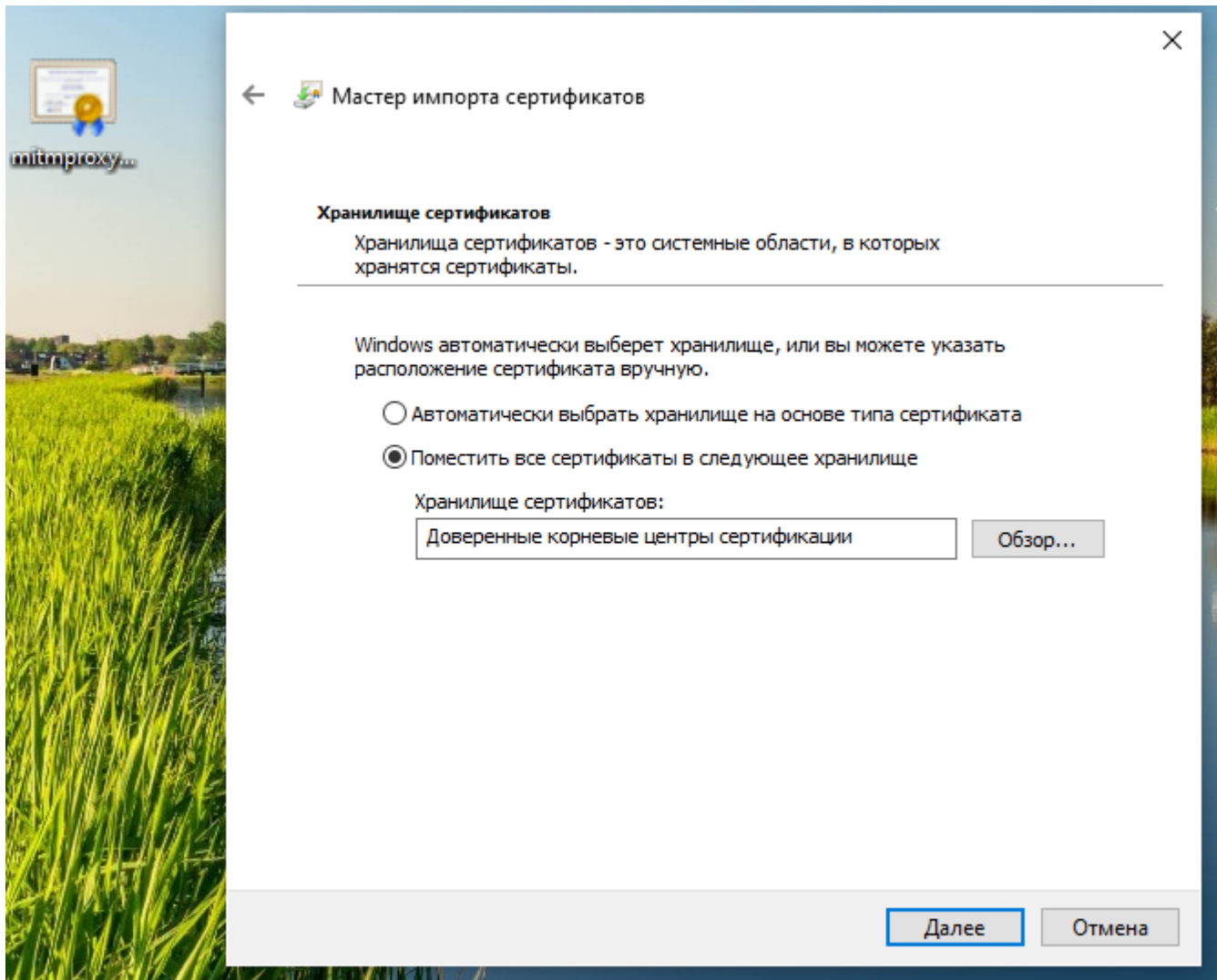


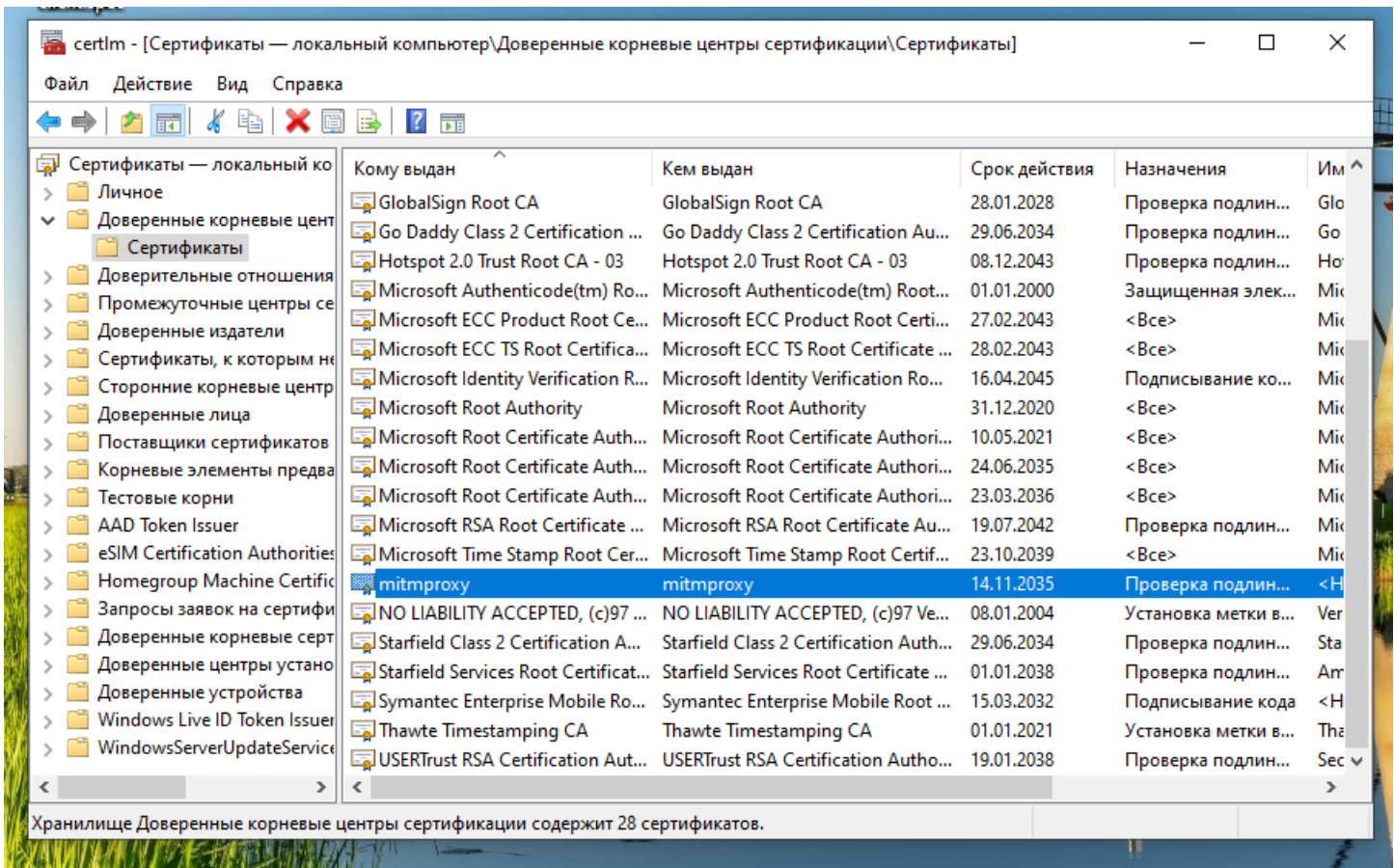
Після завантаження mitmproxу в конфігураційних файлах бачимо сертифікат, перекидую його на вінду-клієнта

```
(kali@kali)-[~/mitmproxy]
$ cd /home/kali/mitmproxy

(kali@kali)-[~/mitmproxy]
$ ls
mitmproxy-ca-cert.cer  mitmproxy-ca-cert.p12
```

Імпорт сертифіката СА mitmproxу до сховища “Довіра до корневих центрів сертифікації” на Windows 7





Додаю правило iptables для перенаправлення вихідного HTTP/HTTPS-трафіку через проксі mitmproxy на Kali Linux

```
(kali@kali)-[~]
$ sudo iptables -t nat -A PREROUTING -p tcp --dport 911 -j REDIRECT --to-port 9999

(kali@kali)-[~]
$ sudo iptables -t nat -L -n

Chain PREROUTING (policy ACCEPT)
target     prot opt source                destination              tcp dpt:911 redir ports 9999
```

Запускаю мітмпроксі

```
(kali@kali)-[~]
$ sudo mitmproxy --mode transparent --showhost --tcp-hosts '*' --set block_global=false -k
```

При відправленні команд MITMProху бачив лише зашифровані блоки даних, Оскільки у моєму варіанті C2-клієнт використовує криптографічну бібліотеку Fernet для шифрування всього трафіку, тому я зміню

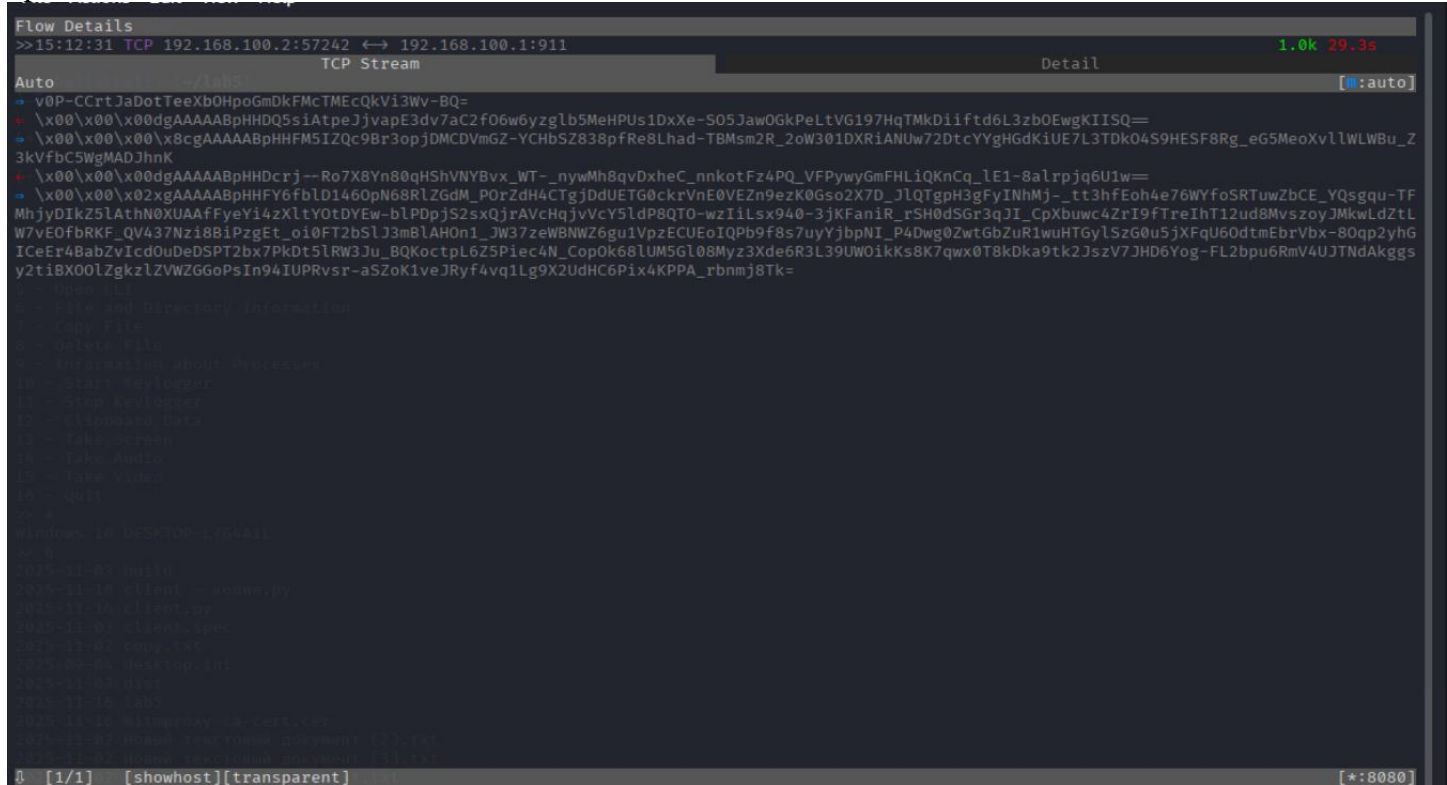
частину коду :
encrypted = cipher.encrypt(data)

s.send(struct.pack("!I", len(encrypted)) + encrypted)

На s.send(struct.pack("!I", len(data)) + data)

return cipher.decrypt(data) на return data

Робота мітпроху до зміни коду: у вікні видно лише зашифровані блоки даних, вміст повідомлень прочитати неможливо.



```
(kali㉿kali)-[~/lab5]  
$ python3 server.py
```

```
[*] Waiting for the connection...
```

```
[+] Success! Connected to client!
```

```
1 - Connect to client  
2 - Close connection  
3 - Shown menu  
4 - Information about System  
5 - Open CLI  
6 - File and Directory Information  
7 - Copy File  
8 - Delete File  
9 - Information about Processes  
10 - Start Keylogger  
11 - Stop Keylogger  
12 - Clipboard Data  
13 - Take Screen  
14 - Take Audio  
15 - Take Video  
16 - Quit
```

```
>> 4
```

```
Windows 10 DESKTOP-L7G4A1L
```

```
>> 6
```

```
2025-11-03 build  
2025-11-16 client - копия.py  
2025-11-16 client.py  
2025-11-03 client.spec  
2025-11-02 copy.txt  
2025-09-04 desktop.ini  
2025-11-03 dist  
2025-11-16 lab5  
2025-11-16 mitmproxy-ca-cert.cer  
2025-11-02 Новый текстовый документ (2).txt  
2025-11-02 Новый текстовый документ (3).txt  
2025-11-02 Новый текстовый документ.txt
```

```
Flow:      Edit      Duplicate      Replay      Export  
>> | :      Help      Back      Events      Options
```

Результат після відключення шифрування: у mitmпроху вже видно текстові команди й відповіді C2-клієнта

```
File Actions Edit View Help
Flow Details
>>15:25:20 TCP 192.168.100.2:57272 ↔ 192.168.100.1:911 508b 15.1s
TCP Stream
Detail
Auto [m:auto]
→ K6oGndu5vkDYWcnRtS36ainUo3a7LuzrdHlPC3goByY=
→ 0000000000 00 00 00 07 73 79 73 69 6e 66 6f ....sysinfo
→ \x00\x00\x00!Windows 10 DESKTOP-L7G4A1L || END ||
→ 0000000000 00 00 00 02 6c 73 ....ls
→ \x00\x00\x01\x962025-11-03 build
→ 2025-11-16 client \xe2\x80\x94 \xd0\xba\xd0\xbe\xd0\xbf\xd0\xb8\xd1\x8f.py
→ 2025-11-16 client.py
→ 2025-11-03 client.spec
→ 2025-11-02 copy.txt
→ 2025-09-04 desktop.ini
→ 2025-11-03 dist
→ 2025-11-16 lab5
→ 2025-11-16 mitmproxy-ca-cert.cer
→ 2025-11-02 \xd0\x9d\xd0\xbe\xd0\xb2\xd1\x8b\xd0\xb9 \xd1\x82\xd0\xb5\xd0\xba\xd1\x81\xd1\x82\xd0\xbe\xd0\xb2\xd1\x8b\xd0\xb9
\x00\xb4\xd0\xbe\xd0\xba\xd1\x83\xd0\xbc\xd0\xb5\xd0\xbd\xd1\x82 (2).txt
→ 2025-11-02 \xd0\x9d\xd0\xbe\xd0\xb2\xd1\x8b\xd0\xb9 \xd1\x82\xd0\xb5\xd0\xba\xd1\x81\xd1\x82\xd0\xbe\xd0\xb2\xd1\x8b\xd0\xb9
\x00\xb4\xd0\xbe\xd0\xba\xd1\x83\xd0\xbc\xd0\xb5\xd0\xbd\xd1\x82 (3).txt
→ 2025-11-02 \xd0\x9d\xd0\xbe\xd0\xb2\xd1\x8b\xd0\xb9 \xd1\x82\xd0\xb5\xd0\xba\xd1\x81\xd1\x82\xd0\xbe\xd0\xb2\xd1\x8b\xd0\xb9
\x00\xb4\xd0\xbe\xd0\xba\xd1\x83\xd0\xbc\xd0\xb5\xd0\xbd\xd1\x82.txt
→ || END ||
→ \x00\x00\x00!Windows 10 DESKTOP-L7G4A1L || END ||
→ 0000000000 00 00 00 02 6c 73 ....ls
→ \x00\x00\x01\x962025-11-03 build
→ 2025-11-16 client \xe2\x80\x94 \xd0\xba\xd0\xbe\xd0\xbf\xd0\xb8\xd1\x8f.py
→ 2025-11-16 client.py
→ 2025-11-03 client.spec
→ 2025-11-02 copy.txt
```

```
Flow Details
>>15:12:31 TCP 192.168.100.2:57242 ↔ 192.168.100.1:911
TCP Stream
Server Connection: /lab5
Address server.py 192.168.100.1:911
Resolved Address 192.168.100.1:911
Client Connection: Connection
Address 192.168.100.2:57242
Timing:
Client conn. established 2025-11-18 15:12:31.675
1 - Server conn. initiated 2025-11-18 15:12:32.331
2 - Server conn. TCP handshake 2025-11-18 15:12:32.332
3 - Client conn. closed -
4 - Server conn. closed -
5 - Open TLI
6 - File and Directory Information
7 - Copy File
8 - Delete File
```

4. Перенесіть реалізацію обробника пакетів з розділу 5.3.5 на Python3 та запустіть на шлюзі з розділу 5.3.2. Модифікуйте трафік Вашого зразку з лабораторної роботи 4. Врахуйте можливість фрагментації пакетів [105].

На Kali Linux використовується два мережеві інтерфейси:

- eth0 — NAT (вихід в Інтернет)
- eth2 — внутрішня мережа (LAN Segment) для зв'язку з Windows

Встановлюється IP-адреса шлюзу:

```
File Actions Edit View Help
(kali㉿kali)-[~]
$ sudo ip addr flush dev eth2
sudo ip addr add 192.168.100.1/24 dev eth2
sudo ip link set eth2 up

[sudo] password for kali:
(kali㉿kali)-[~]
$
```

Вмикаю IPv4-forwarding на Kali Linux для роботи в режимі шлюзу.

```
(kali㉿kali)-[~]
$ echo 1 | sudo tee /proc/sys/net/ipv4/ip_forward

1
(kali㉿kali)-[~]
$
```

Налаштування NAT-маскарадингу: трафік з локальної мережі (eth2) підміняє адресу на інтерфейсі eth0 і виходить в Інтернет через Kali.

```
(kali㉿kali)-[~]
$ sudo iptables -t nat -A POSTROUTING -o eth0 -j MASQUERADE

(kali㉿kali)-[~]
$
```

Windows отримує такі параметри:

IP: 192.168.100.2
Mask: 255.255.255.0
Gateway: 192.168.100.1
DNS: 8.8.8.8

Реалізація обробника пакетів handler.py на Python3: перехоплення TCP-пакетів у NFQUEUE, читання payload і модифікація вмісту.

```
1 #!/usr/bin/env python3
2 from scapy.all import IP, TCP, Raw
3 from netfilterqueue import NetfilterQueue
4
5 TARGET_PORT = 911
6
7 def process(packet):
8     raw = packet.get_payload()
9     ip = IP(raw)
10
11     if ip.frag != 0 or ip.flags.MF:
12         packet.accept()
13         return
14
15     if ip.haslayer(TCP) and ip.haslayer(Raw):
16         data = ip[Raw].load
17
18         print("[*] ORIGINAL:", data[:100])
19
20         new = data.replace(b"sysinfo", b"SYSINFO_HACKED")
21
22         if new != data:
23             print("[+] MODIFIED:", new[:100])
24             ip[Raw].load = new
25             del ip.chksum
26             del ip.len
27             del ip[TCP].chksum
28             packet.set_payload(bytes(ip))
29
30     packet.accept()
31
32
33 def main():
34     nfq = NetfilterQueue()
35     nfq.bind(1, process)
36     print("[*] waiting for packets in NFQUEUE #1 ...")
37     try:
38         nfq.run()
39     except KeyboardInterrupt:
40         pass
41     finally:
42         nfq.unbind()
43
44 if __name__ == "__main__":
45     main()
46
```

Цей код повністю повторює логіку обробника з методички

Далі запускаю клієнта і сервер

Щоб Kali передавала TCP-пакети на порт 911 у обробник Python3, додаємо правила:

iptables у таблиці mangle трафік між клієнтом і сервером на порту 911 пересилається в NFQUEUE №1 для обробки скриптом.

```
(kali@kali)-[~]
$ sudo iptables -t mangle -A PREROUTING -i eth2 -p tcp --dport 911 -j NFQUEUE --queue-num 1
sudo iptables -t mangle -A POSTROUTING -o eth2 -p tcp --sport 911 -j NFQUEUE --queue-num 1
(kali@kali)-[~]
```

Робота обробника пакетів: перехоплення TCP-пакетів C2-клієнта, у терміналі видно оригінальний вміст C2-пакетів (ORIGINAL) та змінений payload (MODIFIED), який відправляється далі

```
(kali@kali)-[~/lab5]
$ sudo python3 handler.py

[*] waiting for packets in NFQUEUE #1 ...
[*] ORIGINAL: b'\x00\x00\x00\x02ls'
[*] ORIGINAL: b'\x00\x00\x00\x07sysinfo\x00\x00\x00\x07cmd dir\x00\x00\x00\x07sysinfo'
[+] MODIFIED: b'\x00\x00\x00\x07SYSINFO_HACKED\x00\x00\x00\x07cmd dir\x00\x00\x00\x07SYSINFO_HACKED'
[*] ORIGINAL: b'\x00\x00\x00\x07sysinfo\x00\x00\x00\x07cmd dir\x00\x00\x00\x07sysinfo'
[+] MODIFIED: b'\x00\x00\x00\x07SYSINFO_HACKED\x00\x00\x00\x07cmd dir\x00\x00\x00\x07SYSINFO_HACKED'
[*] ORIGINAL: b'\x00\x00\x01\x962025-11-03 build\n2025-11-16 client \xe2\x80\x94 \xd0\xba\xd0\xbe\xd0\xbf\xd0\xb8\xd1\x8f.py\n2025-11-16 client.py\n2025-11-03 client.spec'
[*] ORIGINAL: b'\x00\x00\x01\x962025-11-03 build\n2025-11-16 client \xe2\x80\x94 \xd0\xba\xd0\xbe\xd0\xbf\xd0\xb8\xd1\x8f.py\n2025-11-16 client.py\n2025-11-03 client.spec'
[*] ORIGINAL: b'\x00\x00\x01\x962025-11-03 build\n2025-11-16 client \xe2\x80\x94 \xd0\xba\xd0\xbe\xd0\xbf\xd0\xb8\xd1\x8f.py\n2025-11-16 client.py\n2025-11-03 client.spec'
[*] ORIGINAL: b'\x00\x00\x01\x962025-11-03 build\n2025-11-16 client \xe2\x80\x94 \xd0\xba\xd0\xbe\xd0\xbf\xd0\xb8\xd1\x8f.py\n2025-11-16 client.py\n2025-11-03 client.spec'
[*] ORIGINAL: b'\x00\x00\x01\x962025-11-03 build\n2025-11-16 client \xe2\x80\x94 \xd0\xba\xd0\xbe\xd0\xbf\xd0\xb8\xd1\x8f.py\n2025-11-16 client.py\n2025-11-03 client.spec'
[*] ORIGINAL: b'\x00\x00\x01\x962025-11-03 build\n2025-11-16 client \xe2\x80\x94 \xd0\xba\xd0\xbe\xd0\xbf\xd0\xb8\xd1\x8f.py\n2025-11-16 client.py\n2025-11-03 client.spec'
[*] ORIGINAL: b'\x00\x00\x01\x962025-11-03 build\n2025-11-16 client \xe2\x80\x94 \xd0\xba\xd0\xbe\xd0\xbf\xd0\xb8\xd1\x8f.py\n2025-11-16 client.py\n2025-11-03 client.spec'
```

5. Розробіть застосунок, що емулює (sinkhole) сервер керування для Вашого зразку з лабораторної роботи 4,— збирає інформацію про клієнта та подає команду самознищення (зразку, не цільової системи). Розробіть застосунок, що емулює (sinkhole) сервер керування для Вашого зразку з лабораторної роботи 4,— збирає інформацію про клієнта та подає команду самознищення (зразку, не цільової системи).

```
(kali@kali)-[~/lab5]
$ sudo ip addr flush dev eth2
sudo ip addr add 192.168.100.1/24 dev eth2
sudo ip link set eth2 up
echo 1 | sudo tee /proc/sys/net/ipv4/ip_forward

1
```

Реалізація sinkhole-сервера: сервер приймає підключення від C2-клієнта, збирає системну інформацію, список процесів і директорію, а потім надсилає команду quit.

```

1 import socket
2 import struct
3 from datetime import datetime
4
5 PORT = 911
6
7 def send_data(s, data):
8     s.send(struct.pack("!I", len(data)) + data)
9
10 def recv_data(s):
11     raw = s.recv(4)
12     if not raw:
13         return None
14     size = struct.unpack("!I", raw)[0]
15     data = b""
16     while len(data) < size:
17         part = s.recv(size - len(data))
18         if not part:
19             return None
20         data += part
21     return data
22
23 def clean(x):
24     return x.decode(errors="ignore").split(" || END || ")[0]
25
26 def log(x):
27     t = datetime.now().strftime("%Y-%m-%d %H:%M:%S")
28     line = f"[{t}] {x}"
29     print(line)
30     with open("sinkhole_log.txt", "a", encoding="utf-8") as f:
31         f.write(line + "\n")
32
33 sock = socket.socket()
34 sock.setsockopt(socket.SOL_SOCKET, socket.SO_REUSEADDR, 1)
35 sock.bind(("", PORT))
36 sock.listen(5)
37
38 log(f"Sinkhole server listening on port {PORT}")
39
40 while True:
41     conn, addr = sock.accept()
42     conn.settimeout(15)
43     log(f"Client connected: {addr[0]}")
44
45     key = conn.recv(1024)
46     if not key:
47         conn.close()

```

```

47     conn.close()
48     continue
49
50     send_data(conn, b"sysinfo")
51     d = recv_data(conn)
52     if d:
53         log("SYSINFO: " + clean(d))
54
55     send_data(conn, b"proc")
56     d = recv_data(conn)
57     if d:
58         log("PROC: " + clean(d)[:200])
59
60     send_data(conn, b"ls")
61     d = recv_data(conn)
62     if d:
63         log("LS: " + clean(d)[:200])
64
65     send_data(conn, b"quit")
66     log("Sent self-destruct command")
67
68     conn.close()
69

```

Журнал роботи sinkhole-сервера: видно підключення клієнта, отримання sysinfo, переліку процесів і поточного каталогу, а також відправку команди самознищення.

```

(kali@kali)-[~/lab5]
$ python3 sinkhole_server.py

[2025-11-19 00:06:07] Sinkhole server listening on port 911
[2025-11-19 00:06:47] Client connected: 192.168.100.2
[2025-11-19 00:08:41] Client connected: 192.168.100.2
[2025-11-19 00:08:41] SYSINFO: Windows 10 DESKTOP-L7G4A1L
[2025-11-19 00:08:48] PROC:
  PID  0  4  8  12  16  20  24  28  32  36  40  44  48  52  56  60  64  68  72  76  80  84  88  92  96 100 104 108 112 116 120 124 128 132 136 140 144 148 152 156 160 164 168 172 176 180 184 188 192 196 200 204 208 212 216 220 224 228 232 236 240 244 248 252 256 260 264 268 272 276 280 284 288 292 296 300 304 308 312 316 320 324 328 332 336 340 344 348 352 356 360 364 368 372 376 380 384 388 392 396 400 404 408 412 416 420 424 428 432 436 440 444 448 452 456 460 464 468 472 476 480 484 488 492 496 500 504 508 512 516 520 524 528 532 536 540 544 548 552 556 560 564 568 572 576 580 584 588 592 596 600 604 608 612 616 620 624 628 632 636 640 644 648 652 656 660 664 668 672 676 680 684 688 692 696 700 704 708 712 716 720 724 728 732 736 740 744 748 752 756 760 764 768 772 776 780 784 788 792 796 800 804 808 812 816 820 824 828 832 836 840 844 848 852 856 860 864 868 872 876 880 884 888 892 896 900 904 908 912 916 920 924 928 932 936 940 944 948 952 956 960 964 968 972 976 980 984 988 992 996 1000
System Idle Process          0 Services
[2025-11-19 00:08:48] LS: 2025-11-03 build
2025-11-16 client - копия.py
2025-11-16 client.py
2025-11-03 client.spec
2025-11-02 copy.txt
2025-09-04 desktop.ini
2025-11-03 dist
2025-11-16 lab5
2025-11-16 mitmproxy-ca-cert.cer
20
[2025-11-19 00:08:48] Sent self-destruct command

```

файл client.py видалено з файлової системи цільової машини.

Висновки: У ході виконання лабораторної роботи за допомогою INetSim та Cuckoo Sandbox я дослідила поведінку кількох зразків theZoo та побачила, які мережеві запити вони роблять; розгорнула OpenVPN через HTTP-проксі та перевірила, як працює VPN-трафік через проксі-сервер; перенаправила трафік Windows через Kali й перехопила його за допомогою mitmproxу і побачила які дані передає клієнт. Потім я перехопила та змінила пакети через NFQUEUE і Python-обробник, а також підняла sinkhole-сервер, який прийняв дані від клієнта та зміг надати команду на самознищення.